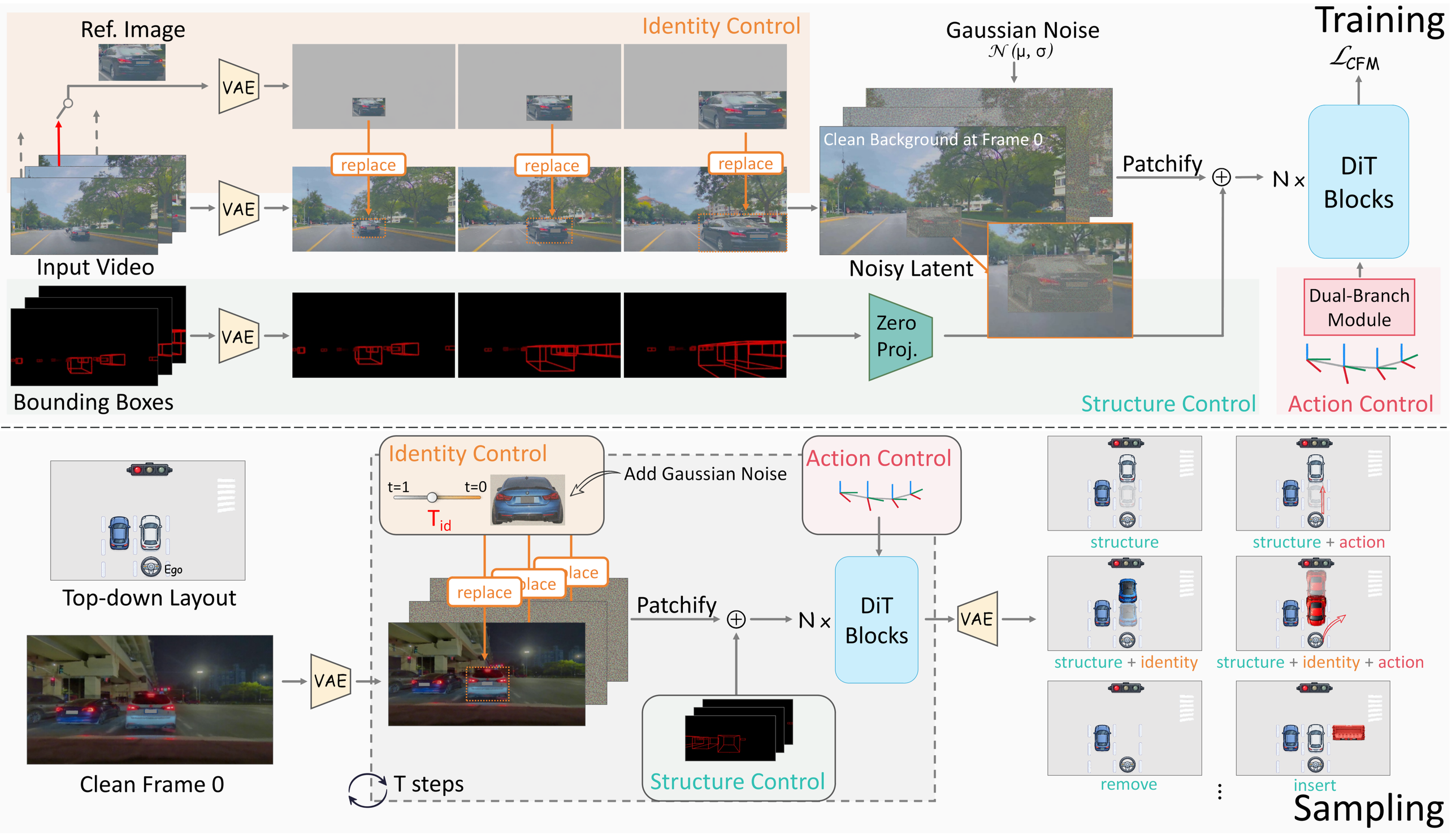
Training
Ref. Image
Identity Control
VAE
replace
Input Video
Gaussian Noise
$\mathcal{N}(\mu, \sigma)$
Clean Background at Frame 0
Patchify
N x
DiT Blocks
$\mathcal{L}_{CFM}$
Noisy Latent
Zero Proj.
Bounding Boxes
Structure Control
Dual-Branch Module
Action Control
Top-down Layout
Ego
Identity Control
t=1
t=0
T_{id}
Add Gaussian Noise
Action Control
Clean Frame 0
T steps
Structure Control
structure
structure + action
structure + identity
structure + identity + action
remove
insert
Sampling